

Table A1 Description of the symbols

Category	Symbol	Description
Index	i, j	General node index; within a_i , b_j , e_i , and e_j , it is used to identify candidate take-off/landing sites, panoramic acquisition points, or general nodes, and in route equations it denotes consecutive ordered nodes.
	g, k	Index of a UAV-feasible connected subregion, $g, k \in K$.
	l	Index of an open take-off/landing site in subregion k ; a_l^k denotes the l th open site in subregion k .
	m	Depending on context, denotes a closed-route index, an initial-solution strengthening operation type, or a summation index over alternative insertion positions in the Regret- γ calculation.
	s, x	s is the normalized position along a take-off/landing-site-to-panoramic-point line-of-sight profile; x is an arbitrary position on a safe horizontal trajectory.
	o, q	o is an ALNS operator index; q is the summation index used in roulette-wheel operator-probability calculations.
	t	Iteration index of G-ALNS-VND / ALNS.
	y	Repair order in Regret- γ ; $y \in \{2, 3, 4\}$.
	u, v	Index u is used in insertion/removal operators to identify the panoramic acquisition point b_u being processed; in the no-fly-zone detour section, u, v denotes a sampled or access point on the no-fly boundary.
	r	Circular coverage radius of a panoramic image; the center spacing between adjacent coverage circles is $3r$ as specified in the text.
Parameter	Ω, H	Administrative boundary of the study area and the no-fly safety exclusion zone.
	K, Ω_k	Index set of UAV-feasible connected subregions and the k th UAV-feasible connected subregion.
	B_o, B, B_k', B_k	Initial full-area panoramic-point set; valid panoramic-point set after no-fly-zone and cultivated land filtering; panoramic-point set in subregion k before line-of-sight filtering; and final valid panoramic-point set after line-of-sight filtering.
	A_o, A, A_k	Original candidate take-off/landing-site set; candidate-site set after filtering by no-fly-zone and valid-DEM conditions; and candidate-site set in subregion k .
	a_i, b_j	The i th candidate take-off/landing site and the j th panoramic acquisition point in a subregion.
	h_i^a, h_j^b	Installation elevation of candidate take-off/landing site a_i and acquisition elevation of panoramic point b_j .

Category	Symbol	Description
	$V_k = [v_{ij}^{ab}]_k$	Candidate-site-to-panoramic-point line-of-sight matrix in subregion k ; $v_{ij}^{ab} = 1$ indicates that the line-of-sight condition is satisfied, otherwise 0 .
	$r_{\max}$	Maximum line-of-sight distance between a take-off/landing site and a panoramic acquisition point.
	$z_{ij}^{ab}(s)$	DEM terrain elevation at normalized position s on the straight-line profile from take-off/landing site a_i to panoramic point b_j .
	$E_k = A_k \cup B_k, e_i$	Unified node set in subregion k and any node in that set.
	h_i	Task elevation of node e_i ; for a take-off/landing site, the corresponding site elevation is used, and for a panoramic point, the acquisition elevation is used.
	L_{ij}^e	Actual safe horizontal trajectory from node e_i to node e_j .
	$z_x, \delta_h, h_{ij}^{\min}$	DEM ground elevation at trajectory position x , terrain-clearance margin, and minimum absolute cruising elevation of segment $e_i \rightarrow e_j$.
	$D_k^{\text{horz}}, D_k^{\text{up}}, D_k^{\text{down}}, D_k$	Horizontal-flight distance matrix, directed ascent-distance matrix, directed descent-distance matrix, and total directed flight-distance matrix in subregion k ; the corresponding matrix elements are $D_{ij}^{\text{horz}}, D_{ij}^{\text{up}}, D_{ij}^{\text{down}}$, and D_{ij} .
	c_l^k, c_r, c_d	Construction cost of open take-off/landing site a_l^k , fixed cost per route, and unit flight-distance cost.
	$1(\cdot)$	Indicator function; equals 1 when the condition in parentheses holds and 0 otherwise.
	M	Maximum number of routes that can be assigned to one open take-off/landing site.
	q_b, Q	Task demand of panoramic acquisition point b and capacity limit of one route.
	$L_{\max}$	Maximum allowable 3D flight distance of a closed route.
	C_i^k	Set of panoramic acquisition points that candidate site a_i in subregion k can serve while satisfying line of sight and the single-point round-trip maximum-distance condition.
	η_m	Minimum relative-improvement threshold for local restructuring operation type m during initial site-route strengthening.
	O_D, O_R, O	Destroy-operator set, repair-operator set, and the operator set currently used in roulette-wheel selection.

Category	Symbol	Description
	$\xi_{\min}, \xi_{\max}, q_{\max}$	Lower and upper bounds of the adaptive destroy rate and the maximum number of panoramic acquisition points that may be removed in one iteration.
	$\theta, w_{\min}$	Reaction factor for operator-weight updating and minimum operator weight.
	T_{end}, α	Terminal simulated-annealing temperature and geometric cooling factor.
	$F_k = [f_{ij}]_k$	No-fly-zone intersection matrix in subregion k ; $f_{ij} = 1$ if the direct horizontal segment between nodes e_i and e_j intersects no-fly zone H , otherwise 0.
	T_i, T_j	Sets of near-tangent boundary access points around nodes e_i and e_j that can be used for no-fly-zone detours.
	$d_{\partial H}(u, v)$	Shortest arc length / shortest boundary-path length along the safe boundary between sampled or access points u and v on the no-fly boundary.
	$D_k^{\text{sh}} = [D_{ij}^{\text{sh}}]_k$	Corrected safe horizontal-distance matrix in subregion k ; stores the actual safe horizontal travel distance for either direct flight or no-fly-zone detours.
	x_i, x_j, x_u, x_v	Planar projected coordinates of nodes e_i and e_j and boundary access points u and v used in the no-fly-zone detour-distance calculation.
	A_k^{open}	Set of candidate take-off/landing sites currently opened in subregion k .
	S_l^k, s_m	Set of closed routes assigned to open site a_l^k and the m th closed route in that set.
	C_k	Total FCLRP cost (objective-function value) in subregion k .
	U_k	Set of panoramic acquisition points not yet covered by the current open-site set during greedy initial site selection.
Variable	$\Delta_u^{\text{ins}}(e_i, e_j), \Delta_u^{\text{new}}(a_i)$	Incremental cost of inserting panoramic point b_u into existing segment (e_i, e_j) , and incremental cost of creating a new single-point route from take-off/landing site a_i .
	X, X'	Current feasible solution and candidate solution during initial-solution strengthening or local restructuring.
	X_t, X_{best}, E	Current solution, historical best solution, and elite pool at ALNS iteration t .
	w_o, p_o	Current weight of operator o and its roulette-wheel selection probability.
	ξ_t, q_t	Actual destroy rate at iteration t and the actual number of panoramic acquisition points removed in that iteration.

Category	Symbol	Description
	Δ_u^{rem}	Flight distance saved by removing panoramic acquisition point b_u in the worst-removal operator.
	$\Delta_{u,(m)}, G_u^y$	The b_u th feasible insertion incremental cost of panoramic point m and its corresponding Regret- y value.
	π_o, n_o	Cumulative reward and number of uses of operator o in the current operator-evaluation segment.
	ΔC_t	Objective-cost difference between the candidate and current solutions at iteration t .
	T_t, P_{acc}	Simulated-annealing temperature and candidate-solution acceptance probability at iteration t .